

5.5 Sense: Exploring Haptic Feedback with Airflow

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Abstract

In this paper, we propose a neckband-type multi-channel airflow presentation device equipped with eight micro blowers. The proposed device employs micro blowers that can be electrically controlled at high speed to achieve a responsiveness on the order of several dozens of microseconds and precise airflow intensity control, thereby enabling delicate and diverse airflow presentation synchronized with video and audio. Additionally, the eight micro blowers arranged around the neck allow a wide range of haptic information, including wind direction, airflow, localized stimulation, and notifications, to be presented with high expressive capability. The findings of this study present new possibilities for airflow presentation as a form of non-contact haptics through experiences that include directional presentation, multi-modal linkage, and interactive operation.

CCS Concepts

• Human-centered computing; • Human computer interaction; • Interaction devices;

Keywords

Interaction devices, Haptic devices, Airflow, Emotional Sensation, Multi-channel

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1 Introduction

Wind is one of the important tactile cues through which humans perceive changes in their surrounding environment. Owing to this characteristic, wind-based haptic feedback has attracted attention as an effective means of enhancing realism and immersion in virtual environments. Haptic presentation devices that use airflow enhance the sense of presence and immersion in virtual environments [1, 2]. Existing presentation devices are mainly classified into wearable devices [1, 3, 5] and fan-type devices placed around displays [4]. However, wearable devices tend to be heavy and impose a large burden on the user when worn for long periods of time, making long-term use difficult. Meanwhile, fan-type devices are positioned far from the user, which often causes delays in airflow arrival and makes fine-grained control challenging.

In this study, we addressed these issues by proposing a neckband-type multi-channel airflow presentation device that uses micro blowers capable of precise electrical control. The device uses a neckband form; therefore, the wearing burden is low, and its light-weight design (less than 150 g) makes it suitable for prolonged use. Furthermore, electric control of the micro blowers enables fine airflow intensity adjustments and presentation timing in synchronization with video and audio, thereby realizing subtle airflow expressions that are difficult with conventional methods.

As a specific example, individual control of the eight micro blowers arranged around the neck enables us to achieve a variety of haptic expressions, such as directional airflow, continuous airflow that conveys a sense of flow, localized stimulation, and notification presentation. These characteristics are expected to enable highly immersive experiences through integration with video and audio, as well as to expand the possibilities for new interaction designs using airflow. We also envision potential use as relaxation devices and accessibility or visual assistance.

2 PROTOTYPE

2.1 HARDWARE DESIGN

The prototype is a neckband-type non-contact haptic device designed to present airflow around the user's neck, and is equipped



Figure 1: (a) Prototype appearance, (b) Blower layout

with eight micro blowers (Figure 1(a)). Each micro blower is arranged such that airflow can be delivered evenly around the neck, and airflow presentation position control is achieved via nozzle angle optimization (Figure 1(b)). The device's external dimensions are 204 mm in length, 190 mm in width, and 38 mm in height, and its weight is approximately 145 g without the cable connected, which reduces the user burden even during long-term use.

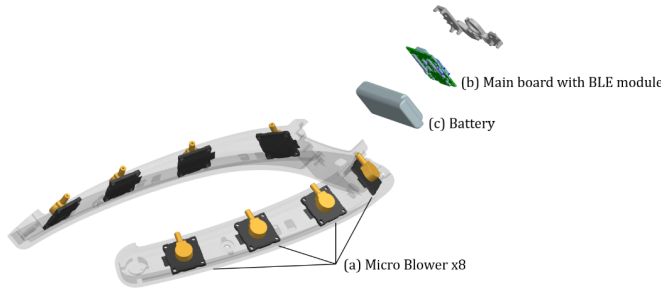


Figure 2: Prototype expanded view

A main board for blower control (Figure 2(b)), a Bluetooth low-energy (BLE) module, and a battery (Figure 2(c)) are integrated in the neckband, and the device operates via either USB wired connection or BLE wireless connection. The blowers can be electrically controlled at high speed on the order of several dozens of microseconds by a microcontroller mounted on the board. The same board is also equipped with an inertial measurement unit (IMU), which enables acquisition of the user's posture information and allows airflow presentation control according to the posture information.

2.2 CONTROL SOFTWARE OVERVIEW

The control software has the following characteristics:

- Individual control of the output of the eight pumps mounted on the neckband is possible.
- The output of each pump can be adjusted in 256 levels.
- The pumps can be controlled in two ways, either directly from a PC using communication commands (Figure 3) or by

registering macros in advance and controlling the device by calling the registered macros.

- The device can acquire posture information (six degrees of freedom (DoF)).
- The PC and neckband are connected via either wired (USB) or wireless (BLE) connection.
- The communication commands define a proprietary human interface device (HID).

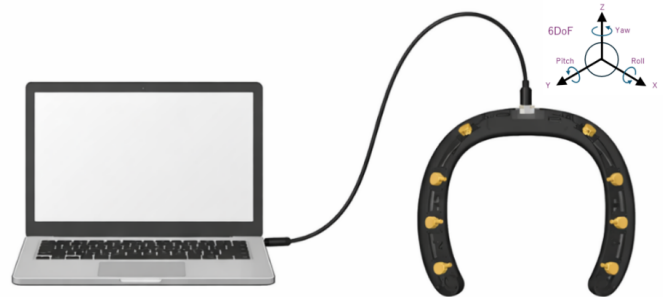


Figure 3: Prototype simply connected to a PC for control

2.3 DEMO CONTENT

In this demonstration, users can experience a variety of haptic expressions that use the characteristics of micro blowers and multi-channel airflow presentation. In each demonstration scene, blower outputs synchronized with video and audio were presented, allowing various haptic patterns to be directly observed and experienced.

Continuous motion creating the impression of flowing wind is expressed by sequentially driving the blowers with shifted timing (Figure 4(a)). New airflow expressions were also explored for notifications and attention cues by switching the blower positions according to direction and changing the presentation duration based on the distance to the target (Figure 4(b)). Furthermore, rapid repetition of strong and weak airflow over a short period was used to represent air fluctuations (Figure 4(c)), which, combined with visuals, achieved sensory expressions such as a sense of presence or

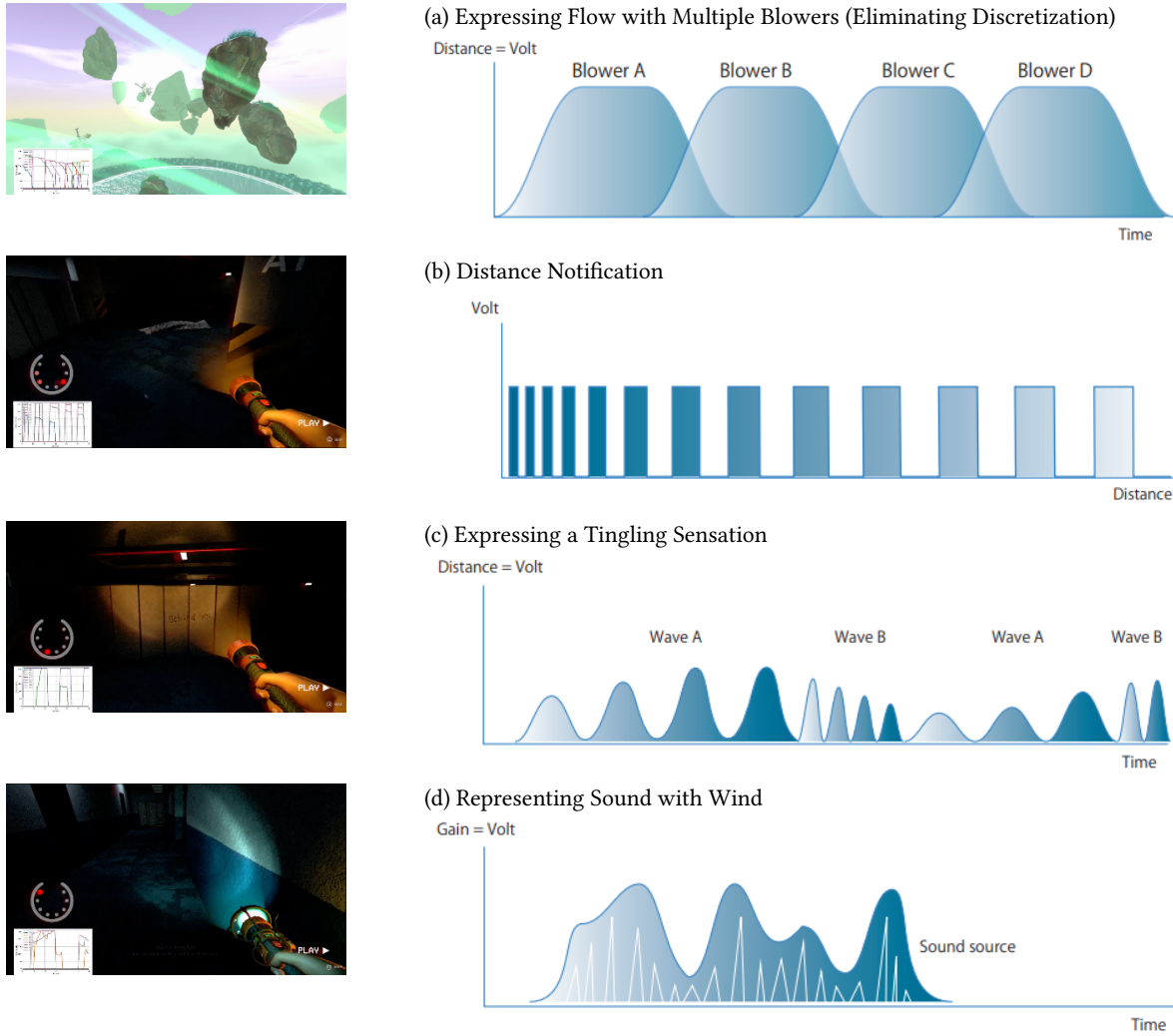


Figure 4: Micro blower expressions

tingling sensations. We also presented airflow in accordance with a human voice (Figure 4(d)) to create a pseudo sense of presence, as if someone was speaking nearby, demonstrating new possibilities for haptic experiences using airflow.

3 CONCLUSIONS

In this paper, we propose a neckband-type multi-channel airflow presentation device that uses micro blowers capable of precise electrical control. The device uses eight blower channels arranged around the neck to address issues encountered in existing wearable and fan-type devices, achieving airflow expressions with excellent directionality and responsiveness. Synchronization with video and audio also enables multi-modal expressions, demonstrating that interaction experiences with a higher level of immersion can be provided. However, due to the characteristics of the micro-blowers, the proposed device has limitations in expressions that require large volumes of airflow.

We demonstrated that fine airflow control offers diverse application possibilities, including environmental expression, object presentation, notification interfaces, and even the creation of psychological effects. In future work, we aim to further expand the possibilities of airflow presentation as non-contact haptics by examining a wider range of interaction scenarios, deepening insights through user evaluations, and combining the proposed system with other virtual- and augmented-reality platforms.

Acknowledgments

The ChatGPT was used to generate Figure 3.

References

- [1] Anke Lehmann, Christian Geiger, Bjorn Woldecke, and Jorg Stocklein. 2009. Design and evaluation of 3D content with wind output. In 3DUI '09: Proceedings of the 2009 IEEE Symposium on 3D User Interfaces, 151–152. <https://doi.org/10.1109/3DUI.2009.4811231>

- [2] Michael Rietzler, Katrin Plaumann, Taras Kranzle, Marcel Erath, Alexander Stahl, and Enrico Rukzio. 2017. VaiR: Simulating 3D airflows in virtual reality. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*, Denver, Colorado, USA, May 2017, 5669–5677. doi: 10.1145/3025453.3026009.
- [3] Yuxuan Sun, Takashi Amesaka, Liwei Chan, and Yuta Sugiura. 2025. AirHook: An ear-based display for simulating the surrounding airflow in virtual reality. In *Proceedings of the 2025 IEEE World Haptics Conference (WHC)*, Suwon, Korea, July 8–11, 2025 (Hands-on Demos)
- [4] Jouke C. Verlinden, Fabian A. Mulder, Joris S. Vergeest, Anna de Jonge, Darina Krutiy, Zsuzsa Nagy, Bob J. Logeman, and Paul Schouten. 2013. Enhancement of presence in a virtual sailing environment through localized wind simulation. In *Faculty of Industrial Design Engineering, Delft University of Technology, Landbergstraat 15, 2628 CE, Delft, the Netherlands* Received 20 March 2013; revised 6 May 2013; accepted 9 June 2013
- [5] Yuichiro Kojima, Yuki Hashimoto, and Hiroyuki Kajimoto. 2009. A novel wearable device to present localized sensation of wind. In *ACE '09: Proceedings of the International Conference on Advances in Computer Entertainment Technology*, 61–65. <https://doi.org/10.1145/1690388.1690399>